

Quantitative Social Structure Data Across Seven Pre-Modern Empires

Bottom Line Up Front: This comprehensive data compilation reveals striking variation in social mobility mechanisms across pre-modern empires, with the Tang examination system achieving **1-2% pass rates** among approximately 2,000 annual candidates, the Ottoman devşirme system processing **200,000-300,000 boys** over three centuries, and Venice maintaining an essentially closed patriciate of **200-250 families** for 500 years. Gini coefficient estimates exist for only four of the seven target empires, ranging from **0.39-0.44 for Rome** to **0.70-0.76 for Venice**, with the Abbasid Caliphate and Mongol Empire lacking any scholarly inequality estimates due to insufficient source materials. The most mathematically formalizable data comes from Venice's Libro d'Oro registrations, Tang examination records, and Ottoman tahrir defters, while nomadic Mongol social structure remains the least quantifiable despite the Yuan dynasty's relatively well-documented four-class hierarchy.

I. Roman Empire (27 BCE – 476 CE Western / 1453 CE Eastern)

Elite composition and senatorial class evolution

The Roman Senate maintained remarkable numerical stability under the Empire following Augustus's reforms. The body standardized at approximately **600 members** after Augustus reduced it from Julius Caesar's inflated 900, maintaining this ceiling through the system of adlectio (direct imperial appointment) used sparingly to fill vacancies. The senatorial order required a property qualification of **1,000,000 sesterces**, (Brainly) (Primidi) necessarily held in Italian land, establishing a wealth threshold roughly **1,111 times** a legionary's annual pay of 900 sesterces.

The equestrian order formed a substantially larger secondary elite, with estimates ranging to approximately **30,000 men** across the empire at its peak in the second century CE. Equestrian status required **400,000 sesterces** in property, (VRoma) creating a two-tier aristocratic structure with meaningful wealth differentiation. Keith Hopkins's foundational quantitative work in *Death and Renewal* demonstrated that despite ideals of hereditary succession, the Roman political elite experienced **high rates of turnover**, with new families regularly entering senatorial ranks while established lineages died out or lost prominence. (Barnes & Noble)

Period	Senate Size	Notes	Confidence
Republic (post-Kings)	300	Standard Republican size	HIGH
Sulla's reforms (80s BCE)	600	Doubled through equestrian inclusion	HIGH
Julius Caesar	900	Dramatic expansion	HIGH
Augustus onward	600	Stabilized; active participation ~100-200	HIGH
Late Empire (4th century)	4,000+	Combined Eastern/Western senates	MODERATE

Citizenship expansion and population structure

Roman citizenship expanded dramatically across the imperial period. The Augustan censuses recorded **4,063,000 citizens** in 28 BCE, rising to **4,937,000** by 14 CE. (Ancient-history-sites) Scholarly debate continues regarding whether these figures represent adult males only (implying substantially larger total citizen populations) or all citizens regardless of age and sex. The *Constitutio Antoniniana* of 212 CE extended citizenship to virtually all free inhabitants of the empire, (Wikipedia) transforming citizenship from a privilege held by perhaps **4-7% of the empire's population** under Augustus to a universal status for free persons.

Walter Scheidel's estimates place the empire's slave population at **10-20% of the total**, (British Museum) with Roman Italy specifically containing **1-1.5 million enslaved persons** representing **13-28% of the Italian population**. (Scholarly Publications) Roman manumission practices were notably more frequent than in Atlantic slave societies, (Trace) with the distinctive feature that freed slaves automatically received Roman citizenship—a provision reflecting and reinforcing the relatively open character of Roman slavery compared to later slave systems.

Gini coefficients and wealth distribution

Branko Milanovic, Peter Lindert, and Jeffrey Williamson's landmark NBER Working Paper 13550 (2007) reconstructed Roman income distribution through social table methodology, (nber) estimating a **Gini coefficient of 39.4** for the empire in 14 CE. (ResearchGate) (IDEAS/RePEc) The **top 1% received 16.1% of total income**, (Munich Personal RePEc Archive) with a cut-off point at **12.4 times mean income**, suggesting a long upper tail of wealth concentration. Scheidel and Friesen's subsequent analysis (2009) placed the Gini slightly higher at **0.41-0.44** for 150 CE, representing the empire at its demographic and territorial peak.

The extraction ratio—actual inequality relative to maximum feasible inequality given subsistence constraints—reached approximately **75-94%** for Roman society, indicating that ancient elites captured far more of the "available surplus" than their modern counterparts. For comparison, contemporary wealthy nations extract only **28-47%** of maximum feasible inequality. (CEPR)

Metric	Value	Period	Source	Confidence
Gini coefficient	39.4	14 CE	Milanovic et al. 2007	MODERATE
Gini coefficient	41-44	150 CE	Scheidel & Friesen 2009	MODERATE
Top 1% income share	16.1%	14 CE	Milanovic et al. 2007	MODERATE
Extraction ratio	75-94%	Imperial period	Milanovic et al. 2007	MODERATE

Social mobility pathways

Roman social mobility operated through multiple channels. Freedmen (*liberti*) faced legal bars from holding office themselves but their children possessed full legal rights, [Wikipedia](#) enabling second-generation advancement to senatorial status. Notable examples include the poet Horace and Emperor Pertinax, both sons of freedmen. The *Augustales*, an elite freedmen guild, provided a status substitute for the magisterial offices forbidden to manumitted slaves.

The *cursus honorum* established a formalized career progression from quaestor through aedile or tribune to praetor and consul. Approximately **20 quaestorships** were filled annually, roughly matching expected vacancies in a 600-member Senate. Senatorial sons possessed automatic eligibility to stand for election, while equestrians required imperial permission for advancement, creating a partially permeable boundary between the two orders.

Data gaps

Precise manumission rates remain unquantified despite clear evidence of their frequency. The PLRE (Prosopography of the Later Roman Empire) provides rich biographical data for 260-641 CE [Amazon](#) but aggregated statistical analyses extracting family turnover rates, elite composition percentages, or systematic mobility metrics from this database remain limited in accessible literature. Cross-class marriage rates, literacy rates by class, and land ownership Gini coefficients lack scholarly estimates.

II. Byzantine Empire (330-1453 CE)

Prosopographical database coverage

Byzantine social history benefits from substantial prosopographical infrastructure. The **PBW (Prosopography of the Byzantine World, 1025-1180)** documents approximately **10,000 individuals** with some **60,000 classified factoids** (data points). The **PBE (Prosopography of the Byzantine Empire)** covers 641-867 CE, while the **PmbZ (Prosopographisches Mittelworgebuch der Byzantinischen Zeit)** encompasses 641-1025 CE in five published volumes. The **Prosopographisches Lexikon der Palaiologenzeit** addresses 1261-1453. Together, these databases provide exceptional coverage of Byzantine elite composition, though systematic statistical analyses remain limited.

The Kletorologion of Philotheos (899 CE) documents **60 higher ranks** in the Byzantine court hierarchy, with notably **50 of these 60 positions available to eunuchs**. Only the imperial throne, praefectus urbi, and quaestor remained closed to castrated officials. Eunuch prominence reflected a distinctive Byzantine mobility pathway: the price differential between a 10-year-old child (**10 nomismata**) and a castrated equivalent (**30 nomismata**) indicates the economic premium on eunuch slaves, with adult eunuchs commanding **50 nomismata** versus **20 nomismata** for intact adult males — [Byzantine World](#) a **150% premium** reflecting anticipated career value.

Dynatoi and aristocratic transformation

Byzantine social structure centered on the distinction between *dynatoi* ("the powerful")—senior civil, military, and ecclesiastical officeholders—and *penetes* (middling and small landowners). Legal poverty was defined as possessing less than **50 gold coins** (*quinquaginta aureos*). The dynatoi remained technically non-hereditary

until the late tenth and early eleventh centuries, when a "limited number of families" monopolized the category, transforming it into an effectively hereditary aristocracy.

The Komnenian period (1081-1185) witnessed a fundamental restructuring of elite composition through the "clan" system connecting families through marriage networks centered on the imperial Komnenos family. Statistical analysis of this network reveals "weakening of the Komnenos family in face of a constantly growing group of affine aristocratic families," demonstrating how marriage alliance patterns altered power distribution within the hereditary elite.

Population and urbanization estimates

Period	Population	Notes	Confidence
Justinian's peak (541)	25-30 million	Before plague	MODERATE
Post-7th century losses	~10-12 million	After Arab conquests; lost 50-60% territory	MODERATE
10th century recovery	~12-18 million	Macedonian dynasty expansion	MODERATE
After 1204	2-4 million	Post-Latin Empire fragmentation	MODERATE
1453	~500,000 or less	Final Byzantine territories	MODERATE

Constantinople's population trajectory illustrates Byzantine demographic volatility: from **300,000-500,000** at Justinian's peak (pre-plague), the city may have declined to as low as **70,000** during the seventh-eighth century "Dark Ages" before partial recovery and eventual decline to approximately **50,000** at the 1453 fall.

Literacy rates in Byzantium present substantial uncertainty but appear higher than Western medieval norms. Lars Brownworth estimates an average of **~30%**, though this "plunged well below 30%" during the seventh-eighth century crisis. Nikolaos Oikonomides's thirteenth-century characterization describes a "completely literate church," "almost completely literate aristocracy," "some literate horsemen," "rare literate peasants," and "almost completely illiterate women." [Wikipedia](#)

Theme system and military-social structure

The theme system that emerged from the seventh-century crisis fundamentally restructured Byzantine social organization. Warren Treadgold estimates **~120,000 thematic soldiers** circa 775 and **100,000-120,000** around 900 CE. The number of themes expanded from an initial **4** (Armeniakon, Anatolikon, Opsikion, Carabisiani) in the 640s-660s to **29** by the tenth century and **38** by the eleventh century's end. Individual theme sizes varied substantially: the Armeniac Theme fielded **9,000-12,000 men** while smaller themes like the Thracesian or Kibyrrhaiotai maintained **4,000-6,000 stratiotai** each.

Military land values established minimum thresholds for stratiotic status: maritime theme sailors required lands worth **4 pounds of gold**, with Constantine VII establishing this as the minimum inalienable value for military

smallholdings.

Wealth distribution and data gaps

No scholarly Gini coefficient estimates exist for the Byzantine Empire in isolation, representing a significant data gap. Milanovic's Byzantine 1000 CE estimate of **Gini 40-45** with the **highest top 1% income share (30.6%)** in his sample of pre-modern societies comes from income reconstruction rather than wealth distribution analysis. (IDEAS/RePEc) (NBER)

Monastic wealth accumulation presents documentation challenges. From the tenth century, monasteries "became even larger and wealthier" through imperial and private donations combined with preferential tax treatment. (World History Encyclopedia) Constantinople contained nearly **30 monasteries** by the mid-sixth century; Mount Olympos in Bithynia housed **50 monasteries**. (World History Encyclopedia) However, precise quantification of monastic landholdings as a share of total land remains unavailable.

[**CONFIDENCE: HIGH** for prosopographical database sizes, court title numbers, and state revenue figures; **MODERATE** for population estimates, military strength, and literacy rates; **LOW/SPECULATIVE** for wealth distribution metrics and precise mobility ratios]

III. Tang Dynasty China (618-907 CE)

Aristocratic clan structure and marriage networks

Tang elite composition centered on the great aristocratic clans organized into four regional groupings: **Northeastern (Hebei) clans** preserving "purest Chinese tradition" and Confucian scholarship; **Northwestern clans** including the ruling Li family with Turkic military heritage; **Northern Shanxi clans** with Central Asian connections; and **Yangzi Valley (Southern) clans** representing Han-era lineages that fled south. The so-called "marriage-ban clans"—families of such prestige that imperial edicts attempted to regulate their intermarriage patterns—included the Cui of Qinghe, Cui of Boling, Lu of Fanyang, Li of Zhaojun, Zheng of Xingyang, and Wang of Taiyuan.

Nicolas Tackett's quantitative analysis in *The Destruction of the Medieval Chinese Aristocracy* (2014) and subsequent work provides the most rigorous statistical treatment of Tang elite composition. His database encompasses approximately **30,000 individuals** reconstructed into **~4,000 patriline**s from **3,329+ tomb epitaphs**. Analysis of marriage networks reveals **87 core patriline**s representing **60% of known elite individuals**, connected through **384 documented marriages**. Two prominent marriage cliques emerge: one centered on the imperial Li clan (Chang'an-based) and one on the marriage-ban clans (Luoyang-based).

Examination system metrics

The Tang examination system provides exceptional documentation for meritocracy metrics. Over the dynasty's 289-year span, **6,585 jinshi degrees** were awarded — (Wikipedia) an average of approximately **23 per year**. Annual candidate pools ranged from **1,000-2,000** for the prestigious jinshi examination, yielding pass rates of **1-**

2% . Wikipedia The less prestigious mingjing examination maintained substantially higher pass rates of **10-20%** , with **100+ candidates selected per examination**.

Metric	Value	Confidence
Total jinshi awarded (entire Tang)	6,585 degrees	HIGH
Average jinshi per year	~23	HIGH
Candidates per year (jinshi)	1,000-2,000	HIGH
Jinshi pass rate	1-2%	HIGH
Mingjing pass rate	10-20%	HIGH

The examination system's importance for elite access increased dramatically over the Tang period. Grand Chancellors with examination backgrounds rose from **~33%** under Emperor Xuanzong (713-756) to **~60%** under Emperor Xianzong (806-821) to **~85%** by the ninth century overall. A 2024 PNAS study analyzing 3,640 male tomb epitaphs confirms that aristocratic family pedigree's effect on career achievement **declined throughout Tang** while examination success **increased in importance**, PNAS with the shift accelerating after Empress Wu's reign (690-705).

Military versus civil elite proportions

The jiedushi (regional military governor) system expanded dramatically following the An Lushan Rebellion (755-763). From **10 frontier commands** in 742 (with approximately **486,900 troops** under their authority), the system grew to **~40 commands** covering the empire by 785. Military participation ratio in 742 stood at approximately **1:103** (soldiers to population), comparable to the late Roman Empire and far lower than early modern European states (France under Louis XIV: 1:50; Prussia 1740: 1:27).

Post-rebellion, the capital elite monopolized significant provincial positions while approximately **75% of circuit governors** came from military backgrounds. The bifurcation between capital-based civil aristocracy controlling the central government and provincially-based military governors dominating regional administration defined late Tang political structure.

Equal-field system and land distribution

The equal-field system (*juntian*) nominally allocated **100 mu** (approximately 16.5 acres or 6.7 hectares) to each adult male aged 18-59, comprising **80 mu** of koufen land reverting to the state upon death and **20 mu** of yongye land permanently inheritable. Elderly, disabled, widows, and young household heads received reduced allocations. The system largely collapsed by the end of the Wu Zhou period (~705) and was formally abandoned in 780 with the adoption of the Two-Tax System, shifting to land- and harvest-based taxation.

No Gini coefficient estimates exist for Tang China. A recent PNAS study provides partial wage inequality measures showing peak inequality during the Tang followed by subsequent decline, but comprehensive wealth or income distribution analysis remains absent from the literature.

Data gaps

Slave population estimates for Tang China remain unavailable despite clear documentation that slaves were purchased along Silk Road routes and comprised substantial proportions of some labor forces—travel party records indicate **2/3 of laborers were slaves belonging to officials** in one excavated sample. Buddhist clergy as an alternative mobility pathway and cross-class marriage rates lack quantitative documentation. Regional examination quotas, specific estate size distributions, and literacy rates by class similarly remain unquantified in accessible scholarship.

[CONFIDENCE: HIGH for examination system statistics; MODERATE for elite composition patterns from Tackett's prosopography; LOW for population structure, wealth distribution, and slave demographics]

IV. Abbasid Caliphate (750-1258 CE)

Ethnic composition transformation of the elite

The Abbasid revolution fundamentally transformed elite composition from Arab dominance under the Umayyads to Persian and eventually Turkish predominance. Under the Umayyads, mawali (non-Arab Muslim converts and their descendants) faced systematic discrimination including taxation with jizya despite their Muslim status. (Oxford Reference) The Abbasids abolished formal mawali discrimination, opening administrative positions to the Persian bureaucratic class. (Islam Reference)

Richard Bulliet's conversion curve methodology, based on Islamic names in biographical dictionaries, provides the best available quantification of this transformation. His estimates for **Iran** show only **~7% Muslim by 750**, rising to **~45% by 800** and achieving majority status by approximately 900. (Encyclopedia.com) (Wikipedia) The al-Zuhrī admission that "knowledge has been taken over by the mawali" reflects their dominance of Muslim scholarly and political positions by the ninth century. (Yaqeen Institute for Islamic Re...)

The Barmakid family exemplified Persian administrative dominance, controlling the vizierate for **17 years** (786-803) under Harun al-Rashid. The kuttāb (secretarial class), described as "largely Persians or Persianized Aramaeans," constituted a distinct social category wielding "considerable political and social influence."

(Encyclopedia.com)

Turkish military elite emergence

Al-Mu'tasim (r. 833-842) reportedly assembled an army of "at least **7,000 Turkic slave soldiers**" (ghulam), a force large enough to cause riots in Baghdad and necessitate relocating the capital to Samarra in 836. (Wikipedia)

(REALM OF HISTORY) The mamluk career trajectory—captured/purchased as children (ages ~12-14), converted to

Islam, trained militarily, manumitted upon completion—created a systematic meritocratic pathway from enslaved outsider to elite military commander or provincial governor. (History Rise)

The price of mamluk boys reached "**three to four times that of a top warhorse**", (History Rise) reflecting the substantial investment in their training and the anticipated return from their service. Notable examples of ghulam advancement include Ahmad ibn Tulun, who rose from ghulam to governor of Egypt (868) and founded the Tulunid dynasty. (Aptigoat)

Bulliet conversion curve estimates

Region	~750 CE	~800 CE	~900 CE	~1000 CE	Confidence
Iran	7%	45%	Majority	~80%	MODERATE
Iraq	—	~40%	—	—	LOW
Egypt	—	—	~50%	—	LOW
Spain	—	—	—	~90%*	MODERATE

*Refers to those who eventually converted; substantial residual populations never converted.

(Taylor & Francis Online)

Conversion accelerated in an S-shaped logistic curve, with early adopters followed by rapid growth and eventual majority establishment. (AlternateHistory.com) The methodology focuses on urban educated males appearing in biographical dictionaries and cannot capture rural populations or provide absolute population percentages with precision.

Population and urban structure

Baghdad at its ninth-century peak housed an estimated **800,000-1,200,000** inhabitants, likely making it the world's largest city until the 930s. (Grokikipedia) (NCESC) The city's urban area covered approximately **7,000 hectares**, "densely populated throughout the eighth-tenth centuries" (Encyclopedia.com) with a "cosmopolitan mix of Arabs, Persians, Berbers, Turks, and non-Muslim dhimmis including Jews and Christians." (Grokikipedia)

Contemporary estimates place the slave population of Iraq in the late ninth century at approximately **~300,000**, with the Basra area specifically containing **~15,000 slaves at any given time** representing approximately **~25% of the labor force**. (Wikipedia) The Zanj Rebellion (869-883) involved "over 50,000 slaves" and cost "tens of thousands of lives." (Wikipedia)

Iqta system and wealth distribution

The iqta system originated in the ninth century as a method to "relieve the state treasury," assigning revenue (not land ownership) to military and civil officials in lieu of salary. (Encyclopedia Britannica) (Islam Reference) Grant-holders (muqti) collected land revenue, maintained troops, and remitted surplus (fawazil) to the central treasury.

Grants were theoretically revocable and non-hereditary with frequent reassignment to prevent local power consolidation, (Wikipedia) though enforcement weakened as central authority declined.

No scholarly Gini coefficient estimates exist for the Abbasid Caliphate, representing a major data gap for mathematical formalization. The Islamic Maghreb's faster ~100-year secular cycles (versus 200-300 years in Western Europe), identified by Turchin building on Ibn Khaldun's framework, suggest distinctive elite dynamics related to higher elite reproduction through polygyny and vulnerability to desert conquest, but these patterns lack direct inequality measurement. (Princeton)

Data gaps

No census data exists for the Abbasid period. No prosopographical database comparable to Roman or Byzantine records has been compiled. Biographical dictionaries (tabaqat) heavily bias toward urban educated male elites. Literacy rates, ulama population counts, artisan/laboring class structures, women's status, and regional variation outside core Iraq/Iran areas all lack quantitative documentation.

[CONFIDENCE: HIGH for institutional structures and elite family histories; MODERATE for Baghdad population ranges, Bulliet conversion patterns, and slave population estimates; LOW for wealth distribution, precise elite proportions, and literacy rates]

V. Ottoman Empire (1299-1922 CE)

Askeri-reaya class proportions

The fundamental Ottoman social division separated the **askeri** (military-administrative class, tax-exempt) from the **reaya** (tax-paying subjects). (Our Times Lines) Reaya comprised **over 90%** of the general population, (DBpedia) with the askeri representing the remaining **less than 10%**, including soldiers, administrators, religious scholars, and anyone granted tax exemption in return for service.

Devşirme system quantification

The devşirme (child levy) system provides exceptional documentation for meritocratic recruitment. Over the system's operational period from the late fourteenth through mid-seventeenth centuries, an estimated **200,000-300,000 boys** were collected, (Islamiqate) with some estimates for the entire system (mid-14th through early 18th centuries) reaching **500,000-1,000,000 non-Muslim adolescent males**.

Metric	Value	Confidence
Total boys collected (15th-17th c.)	200,000-300,000	MODERATE
Collection frequency	Every 4-5 years	HIGH
Collection ratio	1 boy per 40 households	HIGH
Boys per levy	~3,500	MODERATE
Age range	8-18 (typically 8-14)	HIGH
Ideal age	7-10 years	HIGH

Geographic distribution of collections followed documented patterns. The 1603-4 Bursa levy recorded specific numbers from identified regions: Midillu Island provided **105 children**; Gümilcine and surrounding areas provided **168 children**; Siroz region provided **127 children**; Athens/Agriboz region provided **128 children**; Avlonya area provided **145 boys**. [ResearchGate](#)

The Enderun (palace school) trained the most promising recruits for administrative and military careers. **64 Grand Viziers** graduated from this institution. [Daily Sabah](#) Post-1453, Grand Viziers were chosen "nearly exclusively from the kul system"—men of "Byzantine or Balkan background"—after the execution of Çandarlı Halil Pasha ended the previous Turkish family dominance of the position.

Timar system and sipahi counts

Ottoman cavalry organization centered on the timar system, with **37,818 timar holders** recorded in 1525 tax rolls. Treasury records from 1527 document **27,868 timariots** capable of fielding approximately **23,000 cebelus** (retainers). Timar revenue classifications established clear hierarchies:

Classification	Annual Revenue (Akçe)
Timar	Under 20,000
Ziamet	20,000-100,000
Has	Over 100,000

Beylerbeys received **600,000-1,000,000 akçe/year**; sanjakbeys **200,000-600,000 akçe/year**; while ordinary timariot sipahis averaged **2,000-3,000 akçe/year**. In the 1530s, **40% of timariots received less than 3,000 akçe revenue**.

Early timariot ethnic composition shows surprising Christian presence: the 1431 Albania border region contained **16% Christian cavalrymen**, 30% Turks, and 50% Sultan's slaves/Beiks. The 1467-69 Balkans revealed **60% Christian sipahis in Serbia** and **82% in Bosnia** for the first round of timar allocations. Christian sipahis "gradually disappeared due to dispossession or conversion to Islam."

Janissary corps expansion

Period	Numbers	Confidence
1389 (Kosovo)	2,000	MODERATE
Mehmed II (1451-1481)	10,000	MODERATE
1514	16,643	HIGH
Suleiman I	~30,000	MODERATE
1650s	50,000	MODERATE
1680	54,222	HIGH
1694-95	78,798	HIGH
Early 19th century	100,000+	MODERATE

The corps transformed from elite fighting force to urban interest group: by the eighteenth century, nominal strength of **100,000** masked actual combat capability of perhaps **~30,000**, with many "fake" Janissaries holding positions for economic benefits.

Millet system and religious demography

Ottoman population distributed across the millet system with **Muslims comprising approximately two-thirds** of the empire's population, though Turks themselves constituted only **~40%** — a minority in their own empire.

Aithor Istanbul's sixteenth-century religious composition showed approximately **60% Muslims, 30% Christians, and 10% Jews**.

The Jewish population merits particular note: by the end of the sixteenth century, the Ottoman Empire contained the **world's largest Jewish population at 150,000** (compared to 75,000 in Poland and non-Ottoman Ukraine combined). Constantinople and Salonica each housed approximately **20,000 Jews** by the early sixteenth century.

Slave populations in Constantinople during the sixteenth-seventeenth centuries reached approximately **~20% of population**. Wikipedia Total slave imports through Ottoman markets from 1400-1870 may have reached **6-8 million**, including 2 million from Rus'/Polish/Ukrainian sources, 1.5 million from Mediterranean operations, and over 1.3 million from African trade routes. Aleksandarstoyanov

Wealth distribution and guild structure

Evliya Çelebi's seventeenth-century account provides exceptional guild documentation for Istanbul: **158 different artisan associations** with **79,764 esnaf members** operating from **23,214 ateliers and shops**. Large-scale merchants operated **3,188 shops** with 15,160 workers; small-scale merchants ran **14,445 shops** with 48,000 workers. The covered bazaar alone contained **3,000 shops**, with an additional **1,000 shops in caravanserais** and **15,000+ boatmen** forming a powerful corporation.

Ottoman probate records enable wealth distribution analysis. Coşgel and Ergene (2012) and subsequent studies document wealth Gini coefficients of **0.70+** for Ottoman cities, placing them in ranges comparable to Italian commercial cities but derived from different institutional contexts.

Data gaps

Precise Enderun graduation rates to high office remain unquantified. Ulema composition statistics, literacy rates by class and millet, life expectancy differentials, and comprehensive reaya-to-askeri transition rates lack quantitative documentation. Tahrir defter limitations—exclusion of women, children, tax-exempt groups, and nomads—constrain population analysis. [Grokikipedia](#)

[CONFIDENCE: HIGH for Janissary numbers, askeri/reaya classification, devşirme ratios, timar classifications, and Istanbul population ranges; MODERATE for specific devşirme totals, elite composition percentages, slave estimates, and millet proportions; LOW for graduation rates and wealth distribution details]

VI. Venetian Republic (697-1797 CE)

Libro d'Oro and patrician family tracking

Venice offers the most complete elite documentation of any pre-modern state through the Libro d'Oro (Golden Book), officially established by Great Council decree in **1315** [Wikipedia](#) and maintained until the Republic's fall in 1797. The system recorded noble births, marriages, and legitimate offspring, creating an unbroken elite registry spanning nearly five centuries.

Patrician families fell into distinct categories based on antiquity of ennoblement:

Category	Count	Timeframe
Case Vecchie ("Old Houses")	24-25 families	Pre-800 CE; 12 "apostolic" + 4 "evangelical"
Case Nuove ("New Houses")	~150 families	800-1297 CE
Case Nuovissime ("Newest Houses")	30 families	Added after War of Chioggia (1381)
Case Fatte Per Soldo (War of Candia)	~75 families	1645-1669
Case Fatte Per Soldo (Wars of Morea)	~50 families	1684-1718
Final additions	13 families	Late 18th century

Family counts at key dates: **~200 families** at the 1297 Serrata (closure), **207 families** in 1367, **~230 families** in the early eighteenth century, and **~200 families** at the Republic's end.

Great Council membership trajectory

The Great Council (Maggior Consiglio) membership provides time-series data on patrician demographics:

Year	Members	Notes
1143	35	Counsellors
c. 1290	300-400	Pre-Serrata
1297	~950	After Serrata
1315	1,017	—
1340	1,212	—
1527	2,746	Maximum
Mid-16th c.	~2,500	—
Post-1630 plague	1,660	—
1775	1,300	—
1797	~1,100	Final; 547 voted on dissolution

In 1520, **6.4% of adult males** were nobles—a proportion that declined thereafter as the patriciate contracted while the general population stabilized.

Aggregazioni and social closure

The 1297 Serrata dramatically transformed elite access. Pre-Serrata, annual family exit rates from the patriciate ran at **1.2%**; post-Serrata, this dropped to **0.14%** — an order of magnitude reduction in elite turnover. New family entry virtually ceased: between 1297-1381, only **15 families** were added (following the 1310 Tiepolo conspiracy), a period of **84 years** with minimal patriciate expansion.

Subsequent aggregazioni occurred only under crisis conditions:

- **1381**: 30 families (from 62 candidates) after War of Chioggia
- **1645-1669**: ~75 families during War of Candia (**100,000 ducats** each)
- **1684-1718**: ~50 families during Wars of Morea (**100,000 ducats** each)

The 100,000 ducat payment represented extraordinary wealth— "enough to buy a couple or more palaces on the Grand Canal."

Three-tier population structure

Class	Percentage	Functions
Nobili (Patricians)	5-7%	Government, Great Council, all magistracies
Cittadini (Citizens)	7-8%	Bureaucracy, secretarial positions, notaries
Popolani (Commoners)	85-90%	Crafts, labor, trade

The cittadini originarii (original citizens) maintained their own registry, the Libro d'Argento (Silver Book), from 1315. Requirements included birth in Venice as legitimate offspring, no manual labor engagement, census registration without criminal record, and three generations meeting these criteria. The highest office available to cittadini was Cancelliere Grande (Chancellor).

Wealth distribution and Gini estimates

Guido Alfani's research on Venetian terraferma (mainland territories) provides the most rigorous wealth distribution analysis:

Community Type	Average Gini
Cities	~0.90
Agro-towns	~0.85
Rural communities	~0.79

Specific documented values include Bergamo (city) at **Gini 0.715-0.753** across the sixteenth-seventeenth centuries. Venetian territory exhibited the highest pre-modern European inequality levels outside the extreme concentrations found in some Italian city-states.

Alfani's analysis reveals that Venetian taxation was **regressive**: in 1550, the poorest 10% were taxed **5.4-6% of income** while the richest 5% paid only **3.9-4.4%**. By 1750, this disparity widened—poorest 10% taxed **9.3-10.3%**, richest 5% taxed **6.6-7.6%**. This regressive structure drove long-term inequality growth.

The **barnabotti** (impoverished nobles residing around Campo San Barnaba due to cheap rents) represented a distinctive Venetian phenomenon—nobles who retained legal patrician status and Great Council voting rights but lacked wealth to hold unpaid or poorly-paid offices. Government stipends supported them on condition they neither marry nor work, creating a dependent class selling votes in the Broglio.

Doge elections and family concentration

Across the Republic's history, **118 doges** came from only **59 families**. The top 8 families (Contarini, Badoer-Partecipazio, Mocenigo, Sanudo-Candiano, Memmo-Monegario, Morosini, Dandolo, Corner) produced **43 doges (36%)**, indicating substantial concentration within an already closed elite.

[CONFIDENCE: HIGH for Great Council numbers, aggregazioni counts, and family classifications; MODERATE for cittadini numbers, guild statistics, and wealth distribution; LOW for precise literacy rates and life expectancy differentials]

VII. Mongol Empire (1206-1368 CE)

Documentation limitations

The Mongol Empire's nomadic context severely constrains quantitative social documentation compared to sedentary empires. Principal sources include Rashīd al-Dīn's *Jāmi' al-Tawārīkh* (c. 1307), the *Secret History of the Mongols*, and Yuan dynasty Chinese administrative records. The empire's unified period lasted only **54 years** (1206-1260), limiting bureaucratic accumulation, while oral tradition predominated over written documentation and secret burial practices prevented archaeological recovery of Golden Family materials.

Chinggisid lineage structure

The ruling elite centered on the *altan urug* ("Golden Family")—descendants of Genghis Khan and his principal wife Börte. The four primary sons (Jochi, Chagatai, Ögödei, Tolui) and four brothers of Genghis Khan received initial appanage allocations in 1207, establishing sub-lineages that would govern the successor khanates. Only Chinggisids from this legitimate line could rule under the "Chinggisid Principle" that persisted until the nineteenth century (the Kazakh Khanate fell in 1847).

Modern genetic research (Zerjal et al. 2003) suggests approximately **0.5% of the world's male population (~16 million men)** may carry a Y-chromosome lineage potentially linked to Genghis Khan, though this remains speculative. The modern Kazakh Tore clan maintains approximately **~8,000 verified descendants** through documented genealogies.

Keshig (imperial guard) numbers

Period	Size	Confidence
1203 (founding)	150 (70 day + 80 night guards)	HIGH
Early Genghis reign	1,000	HIGH
By 1206 quriltai	10,000 (one tumen)	HIGH
Genghis death (1227)	~10,000	MODERATE
Kublai Khan	12,000	MODERATE

The keshig served as the primary elite formation pathway: "Service in the royal household constituted military and administrative training... the keshig formed the breeding ground for the new elite and future ruling classes." Ordinary keshig soldiers outranked commanders in the regular army, and tümen-ü noyan (myriarchs) were frequently drawn from guard ranks.

Guard composition diversified over time from exclusively Mongol-Turkic to include Chinese, Kipchak, Georgian, Armenian, Alan, Korean, Italian, and Russian units under later khans.

Yuan four-class system

The Yuan dynasty (1271-1368) provides the best-documented Mongol social hierarchy:

Class	Composition	Est. Population %	Privileges
Mongols	Ruling ethnicity	~1.5%	Tax exempt; highest offices; military/judicial privileges
Semu	31 categories of Central/West Asians	~1.5%	Second-rank; finance/tax roles
Hanren	Former Jin subjects (N. Chinese, Jurchens, Khitans, Koreans)	~14%	Third-rank; subordinate positions
Nanren	Former Southern Song subjects	~83%	Lowest rank; barred from high office; heaviest taxation

The 1290 Yuan census recorded approximately **~10 million Hanren** and **~60 million Nanren** for a total of roughly **58 million taxed individuals**. The civil service examination, restored in 1315 after 44 years of suspension, established equal quotas across classes: **75 provincial + 25 metropolitan per class = 300 total jinshi**, though examination difficulty was intentionally lower for Mongols and Semu than for Chinese candidates.

Decimal military organization

Unit	Size	Commander
Arban	10	Arban Jurtchi
Zuun/Jagun	100	—
Mingghan	1,000	Mingghan noyan
Tumen	10,000	Tümen-ü noyan

Tumen counts at key dates: **10-13 tumens (~105,000-130,000 troops)** at the 1206 founding and Genghis's death (1227); **30 tumens (~300,000)** claimed for the Ilkhanate in the 1260s per Rashīd al-Dīn. However, tumens were "often understrength" —nominal 10,000 versus actual average perhaps ~5,000—meaning troop counts cannot be derived directly from tumen numbers.

Population ratios

The fundamental demographic reality of Mongol rule was minority governance: Mongol homeland population circa 1200s numbered approximately **~800,000**, while the empire at peak contained **over 100 million** inhabitants. In Yuan China, Mongols comprised only **~1.5%** of the population, meaning a ratio of roughly **100:1** between ruled and rulers.

Meritocratic advancement examples

Despite the Chinggisid monopoly on supreme rulership, specialized skills enabled non-Mongol and non-elite advancement to high positions:

- **Subutai**: Rose from blacksmith's son to greatest Mongol general
- **Rashīd al-Dīn**: Jewish-born Persian became Ilkhan vizier (1298-1318)
- **Ahmad Fanākātī**: Persian served as Chief Counsellor under Kublai

The ortaq (merchant partnership) system created opportunities for commercial advancement: Mongol princes provided capital while merchants conducted trade and money-lending, resembling Islamic mudharaba or European commenda contracts.

Data gaps

No comprehensive census of noyad (nobles) exists. Slave population estimates remain fragmentary despite clear evidence of systematic capture (one-in-ten boys from captured cities; one-of-three sons enslaved; unmarried adults "legitimate for enslavement"). Hashar (artisan conscription) totals, darughachi appointment records, livestock census data, and wealth distribution metrics lack documentation enabling quantification.

[CONFIDENCE: HIGH for keshig numbers, decimal structure, and four-class categories; MODERATE for Yuan population percentages and army sizes; LOW for total population, slave estimates, and wealth distribution]

Cross-Empire Comparison Tables

Elite Composition Ratios (Hereditary vs. Merit)

Empire	Hereditary Component	Merit Component	Balance Assessment	Confidence
Roman Empire	Senatorial order (1M sesterces qualification); equestrian order hereditary in practice	Freedmen advancement to equestrian; imperial favor	Primarily hereditary with significant mobility through freedmen and imperial patronage	MODERATE
Byzantine Empire	Dynatoi increasingly hereditary from 11th c.; Komnenian clan system	Eunuch advancement (50/60 positions); education pathway; church mobility	Evolved from meritocratic to hereditary over time	MODERATE
Tang Dynasty	Great clans; yin privilege (hereditary protection)	Jinshi examination (1-2% pass rate); ~85% of late Tang chancellors via exam	Most documented shift from hereditary to merit-based; examination dominance by 9th c.	HIGH
Abbasid Caliphate	Arab lineage (declining); Chinggisid principle N/A	Mawali advancement; mamluk system; religious scholarship	Ethnic categories permeable; mamluk system highly meritocratic	MODERATE
Ottoman Empire	Turkish founding families (pre-1453); later hereditary tendencies	Devşirme (200,000-300,000 collected); Enderun school	Classical period highly meritocratic; transformation period increasingly hereditary	HIGH

Empire	Hereditary Component	Merit Component	Balance Assessment	Confidence
Venetian Republic	Libro d'Oro (closed 1297-1797); case vecchie/nuove/nuovissime	Aggregazioni only under crisis (127 families 1645-1718); cittadini bureaucracy	Most closed system; near-total hereditary dominance post-Serrata	HIGH
Mongol Empire	Chinggisid principle (only altan urug rule); noyad status	Keshig advancement; specialized skills (scribes, administrators)	Chinggisid monopoly on rulership; otherwise relatively meritocratic	MODERATE

Gini Coefficient Estimates

Empire	Period	Gini Estimate	Measure Type	Top 1%/10% Share	Confidence	Source
Roman Empire	14 CE	0.39	Income	16.1% (top 1%)	MODERATE	Milanovic et al. 2007
Roman Empire	150 CE	0.41-0.44	Income	~20% (top 1.5%)	MODERATE	Scheidel & Friesen 2009
Byzantine Empire	1000 CE	0.40-0.45	Income	30.6% (top 1%)	MODERATE	Milanovic 2006
Tang Dynasty	618-907	NO ESTIMATE	—	—	—	—
Abbasid Caliphate	750-1258	NO ESTIMATE	—	—	—	—
Ottoman Empire	18th c.	0.70+	Wealth (cities)	—	MODERATE	Coşgel & Ergene 2012
Venetian Republic	1400-1800	0.70-0.90	Wealth	—	HIGH	Alfani & Di Tullio 2019
Mongol Empire	1206-1368	NO ESTIMATE	—	—	—	—

Social Mobility Indicators

Empire	Primary Mobility Mechanism	Quantified Rate	Class Boundary Permeability	Confidence
Roman Empire	Manumission + freedmen advancement	"Frequent" (unquantified); 1-2 generations to full rights	High: slaves→freedmen→citizens→(children) equestrians	MODERATE
Byzantine Empire	Eunuch positions; education; church	50/60 court positions to eunuchs	Moderate: non-hereditary until 11th c.	MODERATE
Tang Dynasty	Jinshi examination	1-2% pass rate; 6,585 degrees over 289 years	Increasing: 33%→85% chancellors via exam	HIGH
Abbasid Caliphate	Mawali integration; mamluk system	Bulliet curve (~7%→45%→majority conversion in Iran)	High for mawali; mamluk to governor common	MODERATE
Ottoman Empire	Devşirme system	1/40 households; 200,000-300,000 total collected	High in classical period (64 Grand Viziers from Enderun)	HIGH
Venetian Republic	Aggregazioni	0.14% annual family exit post-Serrata; 127 families 1645-1718	Very low: essentially closed for 500 years	HIGH
Mongol Empire	Keshig service; specialized skills	10,000 keshig members; examples of advancement	Moderate below Chinggisid level; absolute barrier at top	MODERATE

Class Structure Categories

Empire	Primary Categories	Population Distribution	Legal Distinctions
Roman Empire	Citizens, freedmen, slaves, peregrini	Citizens: 4-7% (pre-212)→all free (post-212); Slaves: 10-20%	Property qualifications (1M/400K sesterces); citizenship rights
Byzantine Empire	Dynatoi, penetes, paroikoi	Dynatoi: small %; ~80% agricultural workers	50 nomismata poverty threshold; tax categories

Empire	Primary Categories	Population Distribution	Legal Distinctions
Tang Dynasty	Liangmin (commoners), jianmin (mean people), slaves	Commoners: majority; hereditary occupational categories	Legal disability for "mean people"; examination eligibility
Abbasid Caliphate	Free Muslims, dhimmis, slaves	Dhimmis: declining (Bulliet curves); slaves: ~300,000 in Iraq	Jizya for dhimmis; conversion = social advancement
Ottoman Empire	Askeri, reaya, kul	Reaya: >90%; Askeri: <10%	Tax exemption; military service obligations
Venetian Republic	Nobili, cittadini, popolani	Nobili: 5-7%; Cittadini: 7-8%; Popolani: 85-90%	Libro d'Oro registration; Great Council membership
Mongol Empire	Mongols, semu, hanren, nanren (Yuan)	Mongols: ~1.5%; Nanren: ~83%	Ethnic quotas in examination; differential taxation

Data Quality Assessment and Gaps

Best-Documented Metrics

The following data points offer the highest confidence for mathematical formalization:

1. Venice Libro d'Oro registrations — Near-complete family and membership counts 1297-1797
2. Tang examination statistics — Pass rates, degree totals, chancellor backgrounds
3. Ottoman devşirme and Janissary records — Collection ratios, corps sizes, tahrir data
4. Roman senatorial and equestrian thresholds — Property qualifications, Senate size
5. Alfani's Venetian wealth Ginis — Archivaly-grounded estimi analysis

Systematic Data Gaps

Category	Empires Lacking Data	Assessment
Gini coefficients	Tang, Abbasid, Mongol	No scholarly estimates exist; mathematically unformalizable
Precise mobility rates	All except Tang examination	Qualitative patterns without percentage quantification
Slave population totals	Tang, Byzantine, Mongol	Fragmentary estimates only

Category	Empires Lacking Data	Assessment
Literacy rates by class	All empires	No reliable pre-modern literacy surveys exist
Life expectancy by class	All except Rome (limited)	Skeletal evidence provides Roman data only
Land ownership Ginis	All empires	Urban wealth distribution better documented than rural land
Cross-class marriage rates	All empires	Documented for elites only (Tackett, Chojnacki)

Methodological Recommendations for Formalization

- Rome and Byzantium:** Social table methodology (Milanovic approach) provides income distribution basis; prosopographical databases enable elite composition modeling
- Tang:** Examination records enable meritocracy modeling; Tackett's patriline database supports network analysis
- Ottoman:** Tahrir defters and devşirme records support recruitment modeling; Janissary time series enables military-social dynamic analysis
- Venice:** Libro d'Oro complete; membership time series directly formalizable; Alfani Ginis applicable
- Abbasid:** Bulliet conversion curves provide S-curve parameters; limited other quantitative basis
- Mongol:** Yuan four-class system proportions available; other metrics speculative

Scholarly Source Summary

Cross-Empire Inequality Studies:

- Branko Milanovic (with Lindert & Williamson): Gini estimates, extraction ratios, inequality possibility frontier
- Walter Scheidel: *The Great Leveler*; Roman income distribution with Friesen
- Guido Alfani: Italian and Venetian wealth inequality; long-term European patterns
- Peter Turchin: Cliodynamics; secular cycles; elite overproduction metrics

Roman Empire:

- Keith Hopkins: *Death and Renewal* — elite turnover

- PLRE (Prosopography of the Later Roman Empire): Biographical database 260-641 CE
- Richard Saller: Family and property
- Peter Garnsey: Social status and legal privilege

Byzantine Empire:

- Alexander Kazhdan: Social change prosopography
- PBW/PBE databases: Prosopographical infrastructure
- John Haldon: Fiscal and military structures
- Warren Treadgold: Theme system military numbers

Tang Dynasty:

- Nicolas Tackett: *Destruction of Medieval Chinese Aristocracy* — quantitative prosopography
- Patricia Ebrey: Aristocratic families
- Denis Twitchett: Cambridge History of China
- PNAS 2024 study: Keju mobility analysis

Abbasid Caliphate:

- Richard Bulliet: Conversion curve methodology
- Roy Mottahedeh: *Loyalty and Leadership*
- Hugh Kennedy: Baghdad and Abbasid political history
- Michael Morony: Iraq social structure

Ottoman Empire:

- Metin Kunt: *Sultan's Servants* — devşirme and provincial administration
- Halil İnalcık: Economic and social history
- Evliya Çelebi: Guild statistics (17th-century primary source)
- Coşgel & Ergene: Ottoman wealth inequality

Venetian Republic:

- Stanley Chojnacki: Marriage and family data
- Guido Alfani: Terraferma wealth distribution

- Frederic Lane: Economic history
- Dorit Raines: Maggior Consiglio statistics

Mongol Empire:

- Thomas Allsen: Administration and merchant partnerships
 - David Morgan: *The Mongols*
 - Elizabeth Endicott-West: Yuan administration
 - Rashīd al-Dīn: *Jāmi' al-Tawārīkh* (primary source)
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Conclusion

This compilation reveals fundamental asymmetries in the quantitative documentation of pre-modern social structures. **Venice** and **Tang China** offer the richest data for mathematical formalization—Venice through its exceptional archival record-keeping producing near-complete elite registries and Alfani's wealth distribution analysis, Tang through its documented examination system enabling precise meritocracy metrics. **Rome** and **Byzantium** benefit from prosopographical infrastructure and Milanovic's income reconstruction methodology, though with wider confidence intervals. **The Ottoman Empire** provides excellent administrative records for the devşirme system and military organization, enabling recruitment and corps evolution modeling.

The Abbasid Caliphate presents the most significant formalization challenge among the sedentary empires—Bulliet's conversion curves offer S-curve parameters, but no Gini estimates, census data, or systematic prosopography exists. **The Mongol Empire** remains least quantifiable due to its nomadic context, short unified duration, and oral tradition dominance, though the Yuan dynasty's four-class system provides some hierarchical proportions.

For CAS Framework Domain 4 mathematical formalization, the compilation suggests prioritizing: (1) Venice for wealth distribution modeling; (2) Tang for meritocracy dynamics; (3) Ottoman for recruitment system formalization; (4) Rome for inequality trajectory analysis. The remaining empires require substantial interpolation from comparative frameworks or acknowledgment of irreducible uncertainty in any quantitative models derived from their social structure data.